

Simone Carletti

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About Me

I am a M.Sc. student in **Computer Engineering** specializing in **Robotics and Autonomous Systems**. I am currently completing my master's thesis at EPFL, developing an adaptive flight control system for a **morphing-wing drone**, giving me hands-on familiarity with UAV dynamics. I have a long-standing passion for **hardware**, having worked extensively with embedded boards like Arduino, RPi, ESP8266, etc. Beyond my academic work, my **professional experience** as a lead developer has allowed me to build strong software engineering skills, focusing on clean and maintainable code. I am proficient in **C++** and **Python**, with extensive experience with **Linux** environments, utilizing **Git**, and **debugging** complex multi-threaded systems.

Education

EPFL - LIS , <i>Master's Thesis</i>	<i>Sept 2025 – Present</i>
◦ Title: $\mathcal{L}_1$-Adaptive augmentation for robust morphing-wing drone flight control ◦ Skills: Control Systems, RL (PPO), C++, Python, Simulink, ROS2, Git, Linux, Unity	
DTU , <i>M.Sc. Autonomous Systems</i>	<i>Sept 2024 – Jul 2025</i>
◦ Erasmus+ exchange - Converted GPA: 30/30	
Politecnico di Torino , <i>M.Sc. Computer Engineering</i>	<i>Sept 2023 – Present</i>
◦ Curriculum in Automation and Intelligent Cyber-Physical Systems - Final GPA: 29.48/30	
Università degli Studi di Firenze , <i>B.Sc. Computer Science and Engineering</i>	<i>Sept 2020 – Sept 2023</i>
◦ Graduated with 110/110 cum laude (with honors)	

Selected Projects

Multiplayer videogame on embedded hardware (GitHub 🔗)	<i>Jan 2024 – Jun 2024</i>
◦ Developed bare-metal C firmware on ARM Cortex-M3 (LandTiger) using Keil μVision ◦ Designed and implemented a custom real-time communication protocol over CAN bus to enable multiplayer support on separate boards. ◦ Ensured deterministic behavior under strict timing and memory constraints.	
Digital Control - Autonomous Robotics (GitHub 🔗)	<i>Oct 2024 – Dec 2024</i>
◦ Designed, simulated, and tuned digital control systems using MATLAB and Simulink to ensure robust mathematical performance before hardware deployment. ◦ Implemented the finalized control algorithms in C/C++ on an embedded microcontroller, successfully bridging the gap between theoretical simulation and real-time execution.	
Localization with EKF in ROS2	<i>Nov 2023 – Dec 2023</i>
◦ Implemented a full EKF localization pipeline from scratch in ROS2 and Python, fusing data from LiDAR, IMU, and wheel odometry	
Autonomous Navigation & Task Execution - DTU RoboCup 🔗	<i>Feb 2025 – May 2025</i>
◦ Developed autonomous navigation, control, and task execution algorithms with continuous integration and validation tests on the real platform	

Experience

Robotics Engineer , <i>DIANA student team</i>	<i>Sept 2023 – Jul 2024</i> <i>Turin, Italy</i>
◦ Collaborated on a Mars rover student project, implementing perception and EKF state estimation in C++ while contributing to software development, testing, and system integration	
Full-Stack Developer , <i>Genomedics S.r.l</i>	<i>Sept 2020 – Present</i> <i>Florence, Italy</i>
◦ Lead developer of three key projects and for the company's main project (KPlatform) ◦ Skills: Angular (Typescript, SCSS, HTML), Vue.js, PHP, Laravel, SQL, Python, Git, Linux	

Skills & Activities

- **Languages:** Fluent in English, Native Italian speaker, Basic Spanish skills
- **Programming:** C++, C, Python, TypeScript, JavaScript, Rust, Java, Matlab, PHP, SQL, HTML, CSS/SCSS, Assembly, LaTeX
- **Tools:** PyTorch, Linux, Git, Google Colab, OpenCV, Onshape, CMake, Docker, Node-RED, Multisim
- **Other:** Raspberry Pi, Arduino, ESP8266, NodeMCU, Soldering, Nvidia Jetson/Orin, Guitar, Photography, Networking, Zigbee